

SHUBHI DIXIT

Aspiring Software Engineer | Second-Year B.Tech CSE Student | AI/ML & Deep Learning | Software Engineering

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EDUCATION

B.Tech (Computer Science Engineering) – Delhi Technological University (DTU)	2025–2029
Class XII (CBSE) – 89.8%	2024–2025
Class X (CBSE) – 95.8%	2021–2022

SKILLS

Languages	C++, Python, JavaScript, TypeScript
Technical	Data Structures & Algorithms, Machine Learning, Deep Learning, LLMs & RAG, OOP, Operating Systems
Tools	React.js, Next.js, Express.js, MongoDB, SQLite, PyTorch, ChromaDB, Git & GitHub

PROJECTS

VCR – Multimodal Visual Commonsense Reasoning | *Python, PyTorch, Deep Learning, Multimodal Learning, SSVL* | [GitHub](#)

- Built and trained a **multi-stage visual reasoning pipeline** with separate **$Q \rightarrow A$** , **$A \rightarrow R$** , and **joint $Q \rightarrow AR$ scorers**, incorporating grounded-image and dual-view representations for answer and rationale prediction.
- Improved task performance from **60% $\rightarrow$ 80% (Q-A)**, **40% $\rightarrow$ 55% (A-R)**, and **15% $\rightarrow$ 40% (Q-AR)** through attention masking, temperature calibration, grounded/dual-view modeling, and targeted training refinements.
- Studied SSVL-based knowledge transfer in depth and investigated Counterfactual Blur, Inverse Rationale Thinking, and Candidate Entailment for stronger visual grounding and rationale consistency.

MuskMelon – Digital Twin of Elon Musk | *Next.js, TypeScript, Temporal RAG, Weaviate Cloud, OpenAI* | [GitHub](#)

- Built an **adaptive, version-controlled knowledge twin** using Temporal RAG, with **Now Mode**, **Time Lens**, **Belief Diff**, and traceable Knowledge Commits for current and historical positions.
- Engineered **context retention, personalization, and grounded LLM orchestration** across Google Drive $\rightarrow$ Weaviate $\rightarrow$ LLM, with Answer Receipts, policy guardrails, and a Knowledge–Voice Firewall.

JeevanMesh – Self-Healing Disaster-Response Drone Swarm | *Next.js, TypeScript, Google Maps API, Distributed Systems, Multi-Agent Simulation* | [GitHub](#)

- Built a **leaderless, partition-tolerant drone-swarm simulation** that autonomously switches scout, verifier, relay and supply roles, transfers low-battery tasks, and continues operating from its last verified mission state during network failure.
- Designed **BubbleNet Adaptive Search + Firefly Relay Positioning**, reducing target-discovery time by **39.7% over grid scanning**, with real-time telemetry, uncertainty mapping, rerouting, and human approval for high-risk actions.

ChronoSync | *ReactJS, Node.js, Express.js, MongoDB, Genetic Algorithm, Greedy Algorithm* | (SIH 2025) | [GitHub](#)

- Built a **web-based timetable scheduling system** for Indian universities, supporting multi-department scheduling, shared classroom/lab allocation, and faculty preference management.
- Designed a **Genetic Algorithm + Greedy Optimization** framework for hard/soft constraint satisfaction, enabling automated timetable generation, conflict detection, schedule regeneration and ERP integration.

NyayaBot | *Python, TypeScript, React, Gemma 4, Ollama, ChromaDB, SQLite* | [GitHub](#)

- Built a **100% on-device, offline legal-assistance engine** on Gemma 4, using Per-Layer Embeddings for inference under 6GB VRAM, with a **hierarchical ChromaDB RAG pipeline** and ShieldAI guardrails for Act $\rightarrow$ Section retrieval, citation verification, and safe outputs.
- Engineered a **ReAct-loop workflow orchestrator** coordinating 8 tools over a 13-table SQLite schema, using **SQL-level compare-and-swap concurrency** and idempotency-key validation for atomic, conflict-safe case operations.

Oon Nirnay — Evidence-first AI decision navigator for rural wool entrepreneurs using LLMs and prompt engineering. [Link](#)

RANKS AND ACHIEVEMENTS

JEE Main: 99.08 percentile | Qualified JEE Advanced
1st Place, *Guessapalooza* | IEEE DTU INVICTUS Annual Technical Fest | ₹3,000 Cash Prize
4th Place, VibeWright Hackathon | Oblivion'26, NSUT | Among 250+ teams
2nd Place, State-Level Mental Mathematics Competition (Class XII) | ₹10,500 Cash Prize
Rank 12, MVPP Delhi State Scholarship Examination (Class IX), among 20,000+ students | ₹5,000 Scholarship
Solved 200+ DSA problems

POSITION OF RESPONSIBILITY

Developer, DTU Times — Contributing to the development and maintenance of DTU Times' web platforms, including frontend/backend features, debugging, and integrations.

EXTRA-CURRICULAR ACTIVITIES

- Volunteered with NSS, mentoring underprivileged children and supporting educational outreach; also participated in debate, poetry, and public-speaking activities throughout school.
- Active participant in technical competitions, hackathons, public speaking, and collaborative team projects across school and university